FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)
BACHELOR IN COMPUTER SCIENCE (HONOURS)
(ODL AND WEEKEND PROGRAMME)

BIT1144/BTL114/BCL1144 PROGRAMMING FUNDAMENTALS

INDIVIDUAL ASSESSMENT

FINAL ASSESSMENT 40%:
Integrated System Development Project

DUE DATE SUBMISSION: 05 APRIL 2026, 11:59PM

STUDENT NAME	
STUDENT ID	

DECLARATION OF INTEGRITY:

I hereby declare that this lab assessment submission is my own independent work and does not contain plagiarized content, unauthorized assistance, or any form of academic dishonesty. I confirm that I have adhered to the academic integrity policies outlined by the university.

I understand that if any form of academic dishonesty, including plagiarism, falsification of results, or unauthorized collaboration, is detected in this submission, I may face disciplinary actions. This may include, but is not limited to, receiving a failing grade for this assessment or further academic penalties as determined by the university's academic integrity committee.

By signing below, I acknowledge my understanding and agreement with this declaration.

Student's Signature: _____ Date: _____

Marking Scheme

Criteria	CLO	Extended Abstract	Relational	Multi-structural	Uni-structural	Pre-structural	Weight	Mark Allocation
System Architecture	CLO2	18–20 Integrated, scalable design	14–17 Logical integration	10–13 Multiple but loosely linked	5–9 Minimal structure	0–4 Poor design	20	
Functional Completeness	CLO2	18–20 Full, robust CRUD	14–17 All required working	10–13 Most working	5–9 Partial	0–4 Fails	20	
GUI Interaction	CLO2	9–10 Intuitive multi-interface	7–8 Functional UI	5–6 Basic UI	2–4 Minimal	0–1 None	10	
Validation & Exception	CLO2	9–10 Robust validation & recovery	7–8 Proper handling	5–6 Basic handling	2–4 Limited	0–1 None	10	
Documentation Quality	CLO3	14–15 Clear, structured, professional	11–13 Good explanation	8–10 Basic report	4–7 Minimal	0–3 None	15	
Reflection & Insight	CLO3	14–15 Deep insight, ethical awareness	11–13 Good reflection	8–10 Basic reflection	4–7 Limited	0–3 None	15	
Ethical Practice	CLO3	9–10 Clear integrity & AI declaration	7–8 Proper declaration	5–6 Minimal	2–4 Incomplete	0–1 Integrity issue	10	
Total							**100**	

CLO ALIGNMENT SUMMARY

CLO	Total Marks	Percentage
CLO2	90	90%
CLO3	10	10%

PERFORMANCE INTERPRETATION SCALE

Mark Range	Performance
80–100	Excellent – Integrated technical and professional competence
65–79	Good – Strong technical with adequate professionalism
50–64	Satisfactory – Functional with limited reflection
40–49	Weak – Incomplete integration
<40	Unsatisfactory

Overall Feedback

First Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date

Second Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date

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Integrated Appointment Booking System - Technical Documentation

Student Name: [Student Name] | **Student ID:** [Student ID]

1. System Overview

This system is a simple Java application for managing client appointments. It supports Regular and Priority appointment types and provides basic GUI features for data entry, storage, and retrieval.

2. OOP Concepts

The system is built on key OOP principles including Inheritance (for the different appointment types), Abstraction (for the base Appointment class), and Polymorphism (for managing a shared list of appointment objects).

Reflection Report: Integrated System Development Project

This project helped me understand how to build a basic Java GUI. One major challenge was managing different layouts and ensuring that buttons functioned as expected.

I also learned about the importance of file handling for data persistence. This process highlighted the need for careful input validation to avoid program crashes.

Source Code Listings

SimplifiedBookingSystem.java

```
import javax.swing.*;
import java.awt.*;
...
public class SimplifiedBookingSystem extends JFrame implements ActionListener { ... }
```

Appointment.java

```
public abstract class Appointment implements Serializable { ... }
```